

Burger stand

- Customers arrive at a burger stand with an average rate of 10 per hour, which can assume to follow an exponential distribution.
- Service times, however, do not follow an exponential distribution. The following sample (each value in minutes) is deemed a true representation of the (service time) population.

5	8	2	7	9	3	2	2	5	6
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- How many people, on average, typically wait in a queue?
- How long do people typically wait in a queue before being serviced?

Burger stand

- Kendall's notation: $M/G/1/GD/\infty/\infty$.
- $L_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}$, where $\rho = \frac{\lambda}{\mu}$
- Call service time distribution **S**
- Let $\mu = \frac{1}{E(S)}$ and $\sigma^2 = \text{var}(S)$, then
- $E(S) = \frac{1}{n} \sum x_i = \mathbf{4.9 \text{ min}}$ and $\text{var}(S) = \frac{1}{n} \sum (x_i - E(S))^2 = \mathbf{6.09 \text{ min}}$
- $\therefore L_q = 2.2803 \text{ people}$, where $\rho = \left(\frac{10}{60}\right) / \left(\frac{1}{4.9}\right)$
- $W_q = \frac{L_q}{\lambda} = 13.68 \text{ min}$

Compressor house

- A compressor house at a steel works has 5 multi-stage compressors, parallel connected, that provide the plants with high-pressure air.
- The steel works can operate without any financial losses if four compressors are in working condition.
- If three compressors are in operation, production losses of R2 million per day are incurred, where these losses rise to R8 million per day under the operation of two compressors.
- One working compressor will result in a loss of R30 million per day, whereas no working compressor will result in damages that translate to a financial loss of R100 million per day.

Compressor house

- A compressor breaks down, on average, once every six months. Repairs take 10 days and there are only two companies in the world that can fly in their technicians to perform the work. You may assume that break downs and repairs follow an exponential distribution.
- What is the expected financial losses to be incurred per annum due to unavailable compressors?
- What is the expected number of compressors waiting in the queue to be serviced?

Compressor house

- A compressor breaks down, on average, once every six months. Repairs take 10 days and there are only two companies in the world that can fly in their technicians to perform the work. You may assume that both break downs and repairs follow an exponential distribution.
- *(M/M/R/GD/K/K)*

Compressor house ($M/M/2/GD/5/5$)

- What is the expected financial losses to be incurred per annum due to unavailable compressors?
- **$Loss = \sum_{i=0}^5 \pi_j \cdot (Cost_j)$, where cost is per 365-day year.**
- $R=2$, therefore, for $j = 0, 1$ and 2 :
- $\pi_j = \binom{K}{j} \rho^j \pi_0$
- $\pi_0 = \binom{5}{0} \left(\frac{2}{36.5}\right)^0 \pi_0 = \pi_0$
- $\pi_1 = \binom{5}{1} \left(\frac{2}{36.5}\right)^1 \pi_0 = 0.273973\pi_0$
- $\pi_2 = \binom{5}{2} \left(\frac{2}{36.5}\right)^2 \pi_0 = 0.030024\pi_0$

Compressor house **(M/M/2/GD/5/5)**

- R=2, therefore, for $j = 3, 4, \text{ and } 5$:
- $\pi_j = \left[\binom{K}{j} \rho^j j! \right] / [R! R^{j-R}] \pi_0$
- $\pi_3 = \left[\binom{5}{3} \left(\frac{2}{36.5} \right)^3 3! \right] / [2! 2^1] \pi_0 = 0.002468 \pi_0$
- $\pi_4 = \left[\binom{5}{4} \left(\frac{2}{36.5} \right)^4 4! \right] / [2! 2^2] \pi_0 = 0.000135 \pi_0$
- $\pi_5 = \left[\binom{5}{5} \left(\frac{2}{36.5} \right)^5 5! \right] / [2! 2^3] \pi_0 = 0.0000037 \pi_0$

Compressor house (*M/M/2/GD/5/5*)

- Remember that $\sum_{i=0} \pi_i = 1$, now solve for π_0 so that:
- $\pi_0 + 0.273973\pi_0 + 0.030024\pi_0 + 0.002468\pi_0 + 0.000135\pi_0 + 0.0000037\pi_0 = 1$
- $\pi_0 = 0.765343$
- $\pi_1 = 0.209683$
- $\pi_2 = 0.022979$
- $\pi_3 = 0.001889$
- $\pi_4 = 0.000103$
- $\pi_5 = 0.00000284$

Loss = R23'526'290.46 per year (remember daily cost must be multiplied with 365 to get annual cost)

Compressor house (**$M/M/2/GD/5/5$**)

- What is the expected number of compressors waiting in the queue to be serviced?
- $L_q = \sum_{j=R}^K (j - R) \pi_j = 0\pi_2 + \pi_3 + 2\pi_4 + 3\pi_5 = 0.002104$

Brewery example

- A local brewery's last two actions on the production line is to cap a beer bottle once it's been filled, followed by labelling the enclosed bottle.
- On average, 200 bottles require to be capped and labelled within any working hour. A single capping machine is available, which can cap an average of 250 bottles per hour when it's not idling.
- There are two labelling machines that operate parallel, and it takes a machine on average 30 seconds to label a bottle.
- Both interarrival times and service times are assumed to be exponential.

Brewery example

- Determine the mean queue length at each machine work-station.
- Determine the total expected time that it takes per beer bottle to undergo the final two production actions.

Brewery example

- $\lambda = 200$ bottles/hour
- $s_1 = 1$
- $\mu_1 = 250$ bottles/hour
- $\lambda < \mu_1$
- $s_2 = 2$
- $\mu_2 = 120$ bottles/hour
- $\lambda < s_2\mu_2 = 2 \cdot 120$
- $\therefore$ Jackson's theorem is applicable.

Brewery example

- For stage 1: $M/M/1/GD/\infty/\infty$
- $\rho = \frac{200}{250}$
- $L_q = \left(\frac{\rho^2}{1-\rho} \right) = \frac{0.8^2}{1-0.8^2} = 3.2 \text{ bottles}$
- $W_q = \frac{L_q}{\lambda} = \frac{3.2}{200} = 0.016 \text{ hour}$

Brewery example

- For stage 2: $M/M/2/GD/\infty/\infty$
- $\rho = \frac{200}{2 \cdot 120}$
- $L_q = \left(\frac{P(j \geq s) \rho}{1 - \rho} \right)$
- $P(j \geq s) = \frac{(s\rho)^s \pi_0}{s!(1-\rho)}$
- $\pi_0 = \left(\sum_{i=0}^{s-1} \frac{(s\rho)^i}{i!} + \frac{(s\rho)^s}{s!(1-\rho)} \right)^{-1} = \left(\frac{\left(2 \cdot \frac{5}{6}\right)^0}{0!} + \frac{\left(2 \cdot \frac{5}{6}\right)^1}{1!} + \frac{\left(2 \cdot \frac{5}{6}\right)^2}{2! \left(1 - \frac{5}{6}\right)} \right)^{-1} = \frac{1}{11}$
- $L_q = \left(\frac{\left(2 \cdot \frac{5}{6}\right)^2}{2! \left(1 - \frac{5}{6}\right)} \cdot \frac{1}{11} \cdot \frac{\frac{5}{6}}{1 - \frac{5}{6}} \right) = 3.7879 \text{ bottles}$
- $W_q = \frac{L_q}{\lambda} = 0.0189 \text{ hour}$
- $\therefore \text{Total time: } 0.016 + 0.0189 = 0.0349 \text{ hour}$

Defects example

- A packaging line sensor at an Electronics Pty(Ltd) can spot two defects in terms of the boxes being used. A defective box can either be folded incorrectly, or don't have a label, or can have both defects at the same time.
- All defective boxes are sent to either a folding station or a labelling station.
- The folding station receives on average 8 boxes per minute from the packaging line, where this station can typically refold a box in 3 seconds.
- The labelling station receives on average 17 boxes per minute and takes on average 2 seconds to perform labelling per box.
- 50% of all boxes that arrive at this folding station requires labelling, even if it just came from the labelling station. The same goes for boxes at the labelling station, but here only 25% requires the folding station.

Defects example

- Assume all inter-arrival and service rates are exponentially distributed, and that both stations have sufficient waiting space.
- What percentage of time is there no boxes to be refolded?
- What is the average number of boxes between both stations?
- What is the expected time a box spends between these two stations?

Defects example

- $r_1 = 8$ boxes/minute,
- $\mu_1 = 20$ boxes/minute,
- $r_2 = 17$ boxes/minute,
- $\mu_2 = 30$ boxes/minute,
- $p_{12} = 0.5$,
- $p_{21} = 0.25$,
- $p_{11} = 0, p_{22} = 0$

Defects example

- $\lambda_j = r_j + \sum_{i=1}^2 p_{ij}\lambda_i$
- $\therefore \lambda_1 = 8 + 0.25\lambda_2 \quad (1)$
- $\therefore \lambda_2 = 17 + 0.5\lambda_1 \quad (2)$
- Solve (1) and (2) simultaneously:
- $\therefore \lambda_1 = 14$ boxes/minute
- $\therefore \lambda_2 = 24$ boxes/minute
- $\lambda_1 < \mu_1$ and $\lambda_2 < \mu_2 \therefore$ Jackson's theorem is applicable.
- Both servers therefore: $(M/M/1/GD/\infty/\infty)$

Defects example

- What percentage of time is there no boxes to be refolded?
- $\pi_0 = 1 - \rho_1 = 1 - \frac{14}{20} = 0.3$. Therefore 30% of the time.
- What is the average number of boxes between both stations?
- $L_1 = \frac{\rho_1}{1 - \rho_1} = \frac{\frac{14}{20}}{1 - \frac{14}{20}} = \frac{7}{3}$
- $L_2 = \frac{\rho_2}{1 - \rho_2} = \frac{\frac{24}{30}}{1 - \frac{24}{30}} = 4$
- $\therefore L = L_1 + L_2 = 4\frac{7}{3}$.

Defects example

- What is the expected time a box spends between these two stations?
- $W = \frac{L}{\lambda} = \frac{L}{r_1 + r_2} = \frac{4\frac{7}{3}}{8+17} = \frac{19}{75}$ minute

I Ching

On the average, 26 patrons per year come to the I.U. library to borrow the *I Ching* (assume that interarrival times are exponential). Borrowers who find the book unavailable leave and never return. A borrower keeps a copy of the *I Ching* for an average of 4 weeks. Let a year comprise 52 weeks.

- If the library has only one copy, what is the expected number of borrowers who will come to borrow the *I Ching* each year and find that the book is not available?
- Suppose that each person who comes to borrow the *I Ching* and is unable to is considered to cost the library \$1 in goodwill. The library wants to procure a second copy, but at the very least the cost should be covered through additional expected goodwill that is received as a direct result thereof within two years. What is maximum cost the library should be willing to pay?

I Ching

- If the library has only one copy, what is the expected number of borrowers who will come to borrow the *I Ching* each year and find that the book is not available?
- $\lambda = 26$ persons/year
- $\mu = 13$ persons/year
- $\rho = \frac{\lambda}{\mu} = \frac{26}{13} = 2$
- $s = 1$
- From Figure 32 on page 1113, obtain a value for $\pi_s = 0.66$. People lost therefore :
- $\lambda\pi_s \approx 26 \cdot 0.66 = 17.16$ persons/year.

I Ching

- Suppose that each person who comes to borrow the *I Ching* and is unable to, is considered to cost the library \$1 in goodwill. The library wants to procure a second copy, but at the very least the cost should be covered through additional expected goodwill that is received as a direct result thereof within two years. What is maximum cost the library should be willing to pay?
- For 2 books:
- $\pi_s = 0.4$.
- $\lambda\pi_s \approx 26 \cdot 0.4 = 10.4$ persons/year
- $\therefore$ additional expected goodwill = $17.16 - 10.4 = \$6.76$ per year.
- $\therefore$ maximum willing price $2 \times 6.76 = \$13.52$.

Exponential inter-arrival times §20.12

- The following inter-arrival times have been observed (in minutes), can an exponential distribution be assumed?

**0.01, 0.07, 0.03, 0.08, 0.04, 0.10, 0.05, 0.10, 0.11, 1.17, 1.50,
0.93, 0.54, 0.19, 0.22, 0.36, 0.27, 0.46, 0.51, 0.11, 0.56, 0.72,
0.29, 0.04, 0.73**

Exponential inter-arrival times §20.12

- χ^2 goodness-of-fit test.
- χ^2 -statistic: $\chi^2(obs) = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}$
 - How are o_i and e_i defined?
 - $5 \leq e_i$ and $k \leq 30$.
 - Value of $\chi^2(obs)$ follows χ^2 -distribution with $DF = k - r - 1 = k - 2$.
- When is distribution assumed to follow exponential one?
 - If $\chi^2(obs)$ is small enough, relative towards your confidence interval α .
 - $\therefore \chi^2(obs) \leq \chi_{k-r-1}^2(\alpha)$, i.e., accept H_0 (null-hypothesis) that assumes exponential inter-arrivals.

Exponential inter-arrival times §20.12

**0.01, 0.07, 0.03, 0.08, 0.04, 0.10, 0.05, 0.10, 0.11, 1.17, 1.50,
0.93, 0.54, 0.19, 0.22, 0.36, 0.27, 0.46, 0.51, 0.11, 0.56, 0.72,
0.29, 0.04, 0.73**

- 25 numbers, thus let $k \leq 5$ (why?). Choose $k = 5$.
- $\bar{\lambda} = \frac{n}{\sum t_i} = \frac{25}{9.19} \approx 2.72$ arrivals per minute.
- Say A is a random variable with density $f(t) = 2.72e^{-2.72t}$.
- $\therefore F(t) = 1 - e^{-2.72t}$ (cdf)

Exponential inter-arrival times §20.12

- 25 observations with $k=5$, thus 20% per group. The following are required to calculate the interval time values in minutes:
- $F(m_1) = 0.2; F(m_2) = 0.4; F(m_3) = 0.6; F(m_4) = 0.8$, where " t " in minute:
- $0 \leq t < m_1$ minute,
- $m_1 \leq t < m_2$ minute,
- $m_2 \leq t < m_3$ minute,
- $m_3 \leq t < m_4$ minute,
- $m_4 \leq t$ minute.

Exponential inter-arrival times §20.12

- Let $t = p$ be any value of allowable " t ", therefore $p = 1 - e^{-2.72t}$
- $\therefore t = \frac{\ln(1-p)}{-2.72}$
- Substitute values for $p = 0.2; 0.4; 0.6$ and 0.8 so that
- $m_1 = \frac{\ln(1-0.2)}{-2.72} = 0.08,$
- $m_2 = \frac{\ln(1-0.4)}{-2.72} = 0.19,$
- $m_3 = \frac{\ln(1-0.6)}{-2.72} = 0.34,$
- $m_4 = \frac{\ln(1-0.8)}{-2.72} = 0.59.$

Exponential inter-arrival times §20.12

- $\chi^2(obs) = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i} = \frac{(6-5)^2}{5} + \frac{(5-5)^2}{5} + \frac{(4-5)^2}{5} + \frac{(5-5)^2}{5} + \frac{(5-5)^2}{5} = 0.40.$
- Choose a confidence interval of 95%.
- $\therefore \chi_3^2(0.05) = 7.81$, so that H_0 can be accepted.